Saussure in HLLSet Algebra

Part I: General Principles

A Computational Exegesis of the *Course in General Linguistics*, Part One, in the Boolean Lattice and GF(2) Ring of HLLSet Algebra

First paper in the series *Saussure in HLLSet Algebra*

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Abstract

The five papers of the EWM PUB series used Saussure's *Course in General Linguistics* as an interpretive lens: langue, parole, the sign, and value were read *into* the HLLSet Boolean lattice and its GF(2) ring. This paper reverses the direction of influence. We read Saussure's text *out of* the algebra: the *Course* is treated as a specification, HLLSet Algebra as the implementation, and each principle of Part One (General Principles) is required to hold either *by construction*—as a theorem of the lattice or the ring—or *by measurement*—as a named invariant of the running system. Part One yields the following dictionary. The linguistic sign is the pair of a lattice node and its materialized context, connected by the two morphisms (μ, ρ); the signifier is a bit-set in the one-dimensional plane, and the signified is the node's differential value in the current frame. Principle I (arbitrariness of the sign) becomes a theorem of coordinate freedom: every permutation of the plane is a simultaneous automorphism of the Boolean lattice and the Boolean ring, so no structural property prefers one IICA morphism over another, while idempotence, immutability, and content addressability fix the chosen bond as the social contract. Principle II (the linear character of the signifier) is realized twice: as the line of bit positions and as GF(2)-linearity of the ring basis. The chapter on immutability and mutability is re-founded as a two-layer theorem: IICA and join-monotonicity make the individual sign immutable, while basis generations re-coordinate the *value* of every sign, so the language changes although no speaker changes it. The chapter on static and evolutionary linguistics becomes a fibration: a synchronic fiber $\mathsf{Syn}(t) = (\mathcal{B}, F(t), \gamma(t))$ over each time step, a diachronic trajectory Dia across fibers, lattice identities as synchronic laws (general but not imperative), recorded transitions as diachronic laws (imperative but not general), the Boolean ring axioms as the panchronic layer that carries no linguistic content, and the chess analogy as the theorem that value is frame-relative. The paper closes with the full dictionary of Part One and a statement of what the reversal of direction changes methodologically.

Keywords: Saussure; Course in General Linguistics; general principles; linguistic sign; arbitrariness; linearity; immutability; mutability; synchrony; diachrony; panchrony; HLLSet; Boolean lattice; GF(2) ring; IICA morphism; basis generation; computational exegesis.

1 Introduction

1.1 The direction of influence, reversed

In the preceding EWM series—the consolidation document, the lattice companion, the state-machine trajectory, the attention traversal, and the closing paper *Langue as a Lattice*—Saussure's *Course in General Linguistics* [1] functioned as an interpretive lens. The *Course* supplied the vocabulary (langue, parole, sign, value, difference) through which the HLLSet Boolean lattice

and its GF(2) ring were recognized as a computable structuralism [5, 6, 7, 8, 9]. The direction of influence ran from Saussure to the algebra.

This paper reverses the arrow. We no longer ask "what does Saussure illuminate in HLLSet Algebra?" We ask: **what does HLLSet Algebra illuminate in Saussure?** The *Course* becomes the object of study, and the algebra becomes the instrument of interpretation. The genre changes accordingly: this is not an engineering paper that borrows a philosophical vocabulary, but a *computational exegesis*—a reading of Saussure's text in which every term must receive a datatype or an operation, and every principle must receive a theorem or an invariant.

1.2 Method: principles as theorems, terms as datatypes

The method has three rules, and we state them in advance because they are the whole discipline of the series.

1. **No metaphor.** A Saussurean term is interpreted only when it has been assigned a concrete object of the algebra (a set, a node, a morphism, a frame, a measure). Until then it remains uninterpreted, and we say so.
2. **Principles become theorems or invariants.** A Saussurean principle is interpreted only when it can be stated as a proposition of the algebra (proved from the definitions) or as an invariant of the running system (a quantity the algebra preserves). If neither is possible, the principle is recorded as a *residual*: an open problem of the interpretation, not a victory.
3. **Ambiguity is data.** Where the *Course* is genuinely ambiguous, the algebra often forces a disambiguation, because a datatype cannot be vague in the way a philosophical term can. We record which disambiguation was chosen, and why the algebra demanded it.

The method is falsifiable in a way that philosophical commentary is not: each Reading below can be checked by running the reference implementation, and each theorem by the usual means.

1.3 Note on the text and the translations

The authors' working copy of the *Course* has been the Russian translation of A. M. Sukhotin, as edited in [4]. For this series, aimed at an English-reading audience, we cite the English translation of Roy Harris [2], with the French original of 1916 [1] as the source of record and the Russian edition retained for the authors' verification. One terminological note is required once. Harris renders *signifiant*/*signifié* as *signal*/*signification*; the older and now-standard English terms, due to Baskin [3], are *signifier*/*signified*. We keep the standard pair and mark the equivalence: *signal* = signifier, *signification* = signified. References to the *Course* are given by part, chapter, and section rather than page numbers, so that they hold across the French, English, and Russian editions.

1.4 What Part One contains, and the series roadmap

Part One of the *Course*, "General Principles," contains three chapters: (I) the nature of the linguistic sign, with the two principles of arbitrariness and linearity; (II) the immutability and mutability of the sign; (III) static and evolutionary linguistics, with the two axes, the two kinds of law, and the chess analogy. This paper follows the same order, after a compact preliminary section that fixes the algebra.

The series is planned to follow the architecture of the *Course* in reverse order of dependency. Part II will treat Synchronic Linguistics (value, "differences without positive terms," syntagmatic and associative relations); Part III, Diachronic Linguistics (phonetic change and analogy as basis pivots and generation drift); Part IV, Geographical Linguistics (dialects and substrata as host languages, codebooks, and the dispatcher); Part V, Questions of Retrospective Linguistics (reconstruction as time travel through old frames). The Introduction of the *Course* (langue and parole, the object of linguistics) was already read in the EWM direction in [9]; its reverse reading

will be folded into Part II, where value requires the langue/parole distinction to be re-founded from the algebra side.

2 Preliminaries: HLLSet Algebra in one page

The reference system is the content-addressed state machine of the EWM programme. We recall only what Part One of the *Course* needs; full treatments are in [5, 6, 7, 8, 9].

Definition 2.1 (Plane, lattice, ring)**.** *The* ***plane*** *is the fixed set*

$$\mathbb{B} \;=\; \{0, \ldots, 32767\}$$

of 32,768 *bit positions. The* ***HLLSet lattice*** *is the powerset*

$$\mathcal{B} \;=\; \mathcal{P}(\mathbb{B}),$$

a Boolean lattice under $\cup$ *(join),* $\cap$ *(meet), and* $\subseteq$ *(order), and simultaneously a Boolean ring under* Δ *(symmetric difference, written* Δ*) and* $\cap$ *[10]. A node of* $\mathcal{B}$ *is an* ***HLLSet****. Cardinalities are written* $|X|$*; the complement of* X *in* $\mathbb{B}$ *is* $\overline{X} = \mathbb{B} \setminus X$*.*

Definition 2.2 (IICA morphism, ingestion, materialization)**.** *A map* $\mu : \mathcal{T} \to \mathcal{B}$ *from tokens to their* ***bit-sets*** *(nodes of the lattice) is an* ***IICA morphism*** *if it is (i)* idempotent*, read as deterministic: the same token always maps to the same bit positions; (ii)* immutable*: once fixed, the image is never rewritten; (iii)* content-addressed*: the image is named by its content alone. The bit positions are* HLLSet-agnostic*: a token sets the same positions whatever HLLSet it occurs in, and an HLLSet is nothing but the union of the positions its tokens set. The morphism extends to finite token sets by union:* $\mu(X) = \bigcup_{t \in X} \mu(\{t\})$*.* ***Materialization*** $\rho : \mathcal{B} \dashrightarrow \mathcal{C}$ *is the partial inverse that restores, from a node, an ordered token context through the door: it reads each set bit and disambiguates, through the LUT fiber of that bit, which tokens set it [8].*

LUT, multi-hashing, and the global gates. Three ideas from the reference system are used throughout the series, and they are introduced here so that the EWM papers need not be read first.

The LUT. Ingestion alone is not reversible: a set bit does not remember which token set it. The **LUT** (look-up table) is the reverse index that restores the memory: each bit position b keeps the fiber $\mathsf{Lut}[b]$—the set of tokens that can set b under the chosen morphism. Materialization reads the set bits of a node and disambiguates through these fibers: exactly on the tokens that built the node, and with collateral candidates only at collided bits.

Multi-hashing: n-grams and n-seeds. One hash is a single, lossy measurement: linear order is lost and collisions are possible. The system therefore measures a stream *several times*, through independent channels: the shared 1-gram channel, windowed n-gram channels (which keep local order as micro-context), and n-seed channels (the same content under different seeds). The idea is not the hash but the *redundancy*: several independent measurements of the same stream, so that what one channel loses, another retains.

The global gates G_1, G_2, G_3. The three canonical channels are realized as three global gates: disjoint regions of the plane (up to hash collisions), one per channel—G_1 the shared default, G_2 the n-gram channel, G_3 the n-seed channel. Any multi-hash HLLSet H is decomposed by intersecting it with the gates,

$$H \;=\; (H \cap G_1) \;\sqcup\; (H \cap G_2) \;\sqcup\; (H \cap G_3),$$

mutually exclusive up to hash collisions: as regions of the plane the gates are disjoint, $G_i \cap G_j = \varnothing$ for $i \neq j$, and only a collided token can set a bit in an unexpected gate. This decomposition

is the workhorse of materialization and of the bridge between worlds in Part IV: each part of H carries a different channel's witness, and the parts can be read separately because they are disjoint.

Proposition 2.3 (Ingestion is join-preserving; the morphism step)**.** *For finite token sets $X, Y \subseteq \mathcal{T}$,*

$$\mu(X \cup Y) \;=\; \mu(X) \cup \mu(Y).$$

Consequently the statistics of usage (co-occurrence, overlap, containment) reduce to one operation, union, on one shared plane; frequency and linear order are forgotten at the door and restored, when needed, by materialization. (Throughout this paper μ is the flat, join-preserving IICA morphism; the windowed n-gram morphism of [7] satisfies the inclusion form $X \subseteq Y \Rightarrow \mu(X) \subseteq \mu(Y)$, and the linguistic readings below hold in the image of whichever IICA morphism is chosen.)

Definition 2.4 (BSS, document coverage, basis frames, generations)**.** *Against a reference node $B_w \subseteq \mathbb{B}$, the **directed soft key** is*

$$\mathsf{BSS}_w(X) \;=\; \frac{|X \cap B_w|}{|B_w|}.$$

*For a context X read against a document D, the **document coverage reading** is*

$$N_D = |D \setminus X|, \qquad R_D = |D \cap X|, \qquad \mathrm{cov} = R_D/|D|.$$

*A **basis frame** is a reduced (RREF) GF(2) basis $F = (B_1, \ldots, B_k)$ of a span of observed nodes; the frame is stamped with a **generation** γ that increments whenever the basis is recomputed. Relative to F, a node $X \in \langle F \rangle$ has a unique **hard key** $c \in \mathrm{GF}(2)^k$ with*

$$X \;=\; \bigoplus_{i:c_i=1} B_i,$$

*the **soft key** $w_i = \mathsf{BSS}_{B_i}(X)$ for $i = 1, \ldots, k$, and the **spill** $s = |X \setminus \mathrm{cover}(F)|$ measuring what the frame cannot yet see [7].*

Remark 2.5 (Permutation invariance)**.** *Every invariant used in this paper—cardinalities, BSS, the document coverage numbers, rank, spill—depends only on cardinalities of intersections and unions. Hence it is unchanged by any permutation of the plane. This single observation will carry the formal reading of arbitrariness in Section 3.*

3 Chapter I: The nature of the linguistic sign

3.1 Sign, signifier, signified

Saussure opens Part One by rejecting the nomenclature view: a language is not "a list of terms corresponding to a list of things" [2, Part I, ch. I, §1]. The linguistic sign is a *two-sided psychological entity* uniting, not a thing and a name, but a concept and a sound pattern. The two sides are inseparable, like the two faces of a sheet of paper; and both are psychological—the sound pattern is not the material sound but its imprint on the speaker's mind.

The HLLSet reading drops the psychologism and keeps the structure: a sign is a *two-sided object* in which the two sides are connected by two morphisms, not by a sheet of paper.

Reading 3.1 (The linguistic sign). Let $A \in \mathcal{B}$ be a node, $\mathcal{C}_A = \rho(A)$ its materialized context, and $X_A \subseteq \mathcal{T}$ the token content that ingests to A, so that $\mu(X_A) = A$. The **sign** is the pair

$$\mathcal{S} \;=\; (\mu, \rho), \qquad \mu : \mathcal{C} \to \mathcal{B}, \qquad \rho : \mathcal{B} \dashrightarrow \mathcal{C},$$

together with the two faces of any node:

$$\begin{array}{ccc} \mathcal{C} & \xrightarrow{\mu} & \mathcal{B} \\ & & \downarrow{\scriptstyle \rho} \\ & & \mathcal{C} \end{array}$$

- The **signifier** (sound pattern) is the node A itself: a bit-set in the one-dimensional plane, a pattern in a fixed medium, exactly as the acoustic image is a pattern in the fixed medium of time.
- The **signified** (concept) is the *differential value* of A: its hard key $c \in \mathrm{GF}(2)^k$ in the current frame, its complement $\overline{A}$, and—on the side of usage—the context $\mathcal{C}_A$ that materializes from it.
- The **bond** between the two faces is the pair of morphisms (μ, ρ), with the round-trip guarded by coverage: $\rho(\mu(\mathcal{C})) = \mathcal{C}$ up to order iff $\mathrm{cov}(\mathcal{C}; D) = 1$, where the coverage is read against the document D from which the context $\mathcal{C}$ was drawn [9].

Two comments on the reading. First, the signifier is assigned to the node, not to the token, because the node is what is *shared* across hosts and across paroles; the token is the private currency of one host. Second, the signified is assigned to the differential value, not to a "meaning," because Saussure himself defines value negatively—a sign is what the others are not—and the algebra makes that definition executable as a coordinate vector in a basis. Full treatment of value is the subject of Part II; here only its role as the second face of the sign is needed. Third, Saussure's own examples are words, and a word is a token; the reading must not collapse the two. The signifier is the HLLSet node that the token collection ingests to—even when that node is a single activated bit. A token is parole's currency; a node, however small, is langue's.

Observation 3.2 (Two-sidedness without psychologism). Saussure's insistence that both faces are psychological becomes, in the algebra, the statement that both faces are *structural*: the signifier is a node of the shared lattice (not the physical sound), and the signified is a position in the frame (not a private mental content). The sheet of paper is replaced by the requirement that the two morphisms be adjoint on the image, so that neither face exists without the other [9].

3.2 Principle I: The arbitrariness of the sign

Saussure's first principle: "The link between signifier and signified is arbitrary" [2, Part I, ch. I, §2]. He immediately adds two clarifications that the later literature often loses. First, *arbitrary* does not mean that the signifier depends on the free choice of the speaker; it means *unmotivated*: there is no natural connection between the signifier and the signified. Second, the whole linguistic system rests on this irrational principle, and it is precisely arbitrariness that makes the system a matter of *convention* rather than of nature.

The HLLSet reading has three layers: arbitrariness as coordinate freedom, arbitrariness as the freedom of the morphism, and fixedness as IICA.

Reading 3.3 (Arbitrariness). The link between signifier and signified is the IICA morphism μ together with its materialization inverse. The link is arbitrary in the following exact sense: *nothing in the content of a token selects its node.* The reference implementation chooses a hash, and a seed for it, for the single reason that it spreads values evenly over the plane; any other deterministic content-addressed map would produce the same kind of system. The link is *unmotivated* because the Boolean algebra cannot see which map was chosen.

The formal content of that Reading is a theorem.

Theorem 3.4 (Arbitrariness as coordinate freedom)**.** *Let $\pi : \mathbb{B} \to \mathbb{B}$ be any permutation of the plane, and let*

$$\pi_* : \mathcal{B} \to \mathcal{B}, \qquad \pi_*(A) \;=\; \{\pi(i) : i \in A\}.$$

Then π_* is simultaneously an automorphism of the Boolean lattice $(\mathcal{B}, \cup, \cap)$ and of the Boolean ring $(\mathcal{B}, \Delta, \cap)$. Consequently no structural property of the algebra—BSS, the coverage readings, rank, spill, hard key cardinality—can distinguish the ingestion morphism μ from the re-coordinated morphism $\pi_* \circ \mu$: every such invariant is unchanged, provided the reference objects (the basis, the document, the reference nodes) are re-coordinated along with the nodes, i.e. read through the same relabeling of the plane.

Proof. A permutation preserves cardinality and commutes with union, intersection, and complement: $\pi_*(A \cup B) = \pi_*(A) \cup \pi_*(B)$, and likewise for $\cap$ and $\setminus$. Hence all cardinality-defined invariants are preserved when their reference objects are re-coordinated too. The lattice and ring identities are preserved because π_* is a bijective homomorphism. Thus the two systems, μ and $\pi_* \circ \mu$, are isomorphic; the theory cannot prefer one. □

Corollary 3.5 (Unmotivated, not undetermined)**.** *Arbitrariness in Saussure's sense is* coordinate freedom*, not indeterminacy. The bond is unmotivated because any coordinate system works; it is nevertheless* fixed*, because a community that wishes to communicate must share one. In the algebra, the fixedness is the IICA contract itself: idempotence makes the bond deterministic, immutability makes it stable, and content addressability makes it the same for every host. Arbitrary yet fixed: coordinate freedom plus IICA.*

Remark 3.6 (The seed as the social contract)**.** *The reference implementation exposes the fixedness in a concrete form: the hash* type *and its* seed *are parameters of the measurement regime, not of the lattice [6]. Two communities with different seeds are two languages over the same plane; within one community, the seed is never up for individual renegotiation. This is Saussure's "the individual cannot choose" stated as an access-control property of the system.*

3.3 Principle II: The linear character of the signifier

Saussure's second principle: the signifier, being auditory, unfolds in time and therefore (a) represents a span, and (b) the span is measurable in a single dimension—it is a line [2, Part I, ch. I, §3]. Visual signifiers, by contrast, can offer groupings in several dimensions at once. The whole mechanism of language, Saussure concludes, depends on this linearity.

The HLLSet reading realizes the line twice.

Reading 3.7 (Linearity of the signifier). The signifier $A \subseteq \mathbb{B}$ is a subset of the *one-dimensional* medium $\mathbb{B} = \{0, \ldots, 32767\}$: a line of bit positions. Ingestion takes the temporal line of speech—the chain of signifiers in parole—and compiles it onto this spatial line, forgetting order; materialization restores temporal order at the door. The same object is *linear* in the algebraic sense: $\mathcal{B}$ is a vector space over GF(2) of dimension $|\mathbb{B}|$, and relative to a frame F every node in the span has a unique coordinate vector. The one-dimensionality Saussure required of the signifier is therefore realized twice: as the line of bit positions, and as GF(2)-linearity of the ring.

Proposition 3.8 (The signifier is linear in two senses)**.** *(i)* Medium linearity. *Every signifier is a subset of the single ordered line $\mathbb{B}$; no signifier needs a second dimension to be written. (ii)* Algebraic linearity. *For every basis frame $F = (B_1, \ldots, B_k)$ and every node $X \in \langle F \rangle$, there is a unique $c \in \mathrm{GF}(2)^k$ with*

$$X = \bigoplus_{i : c_i = 1} B_i.$$

The hard key c is the signifier's address in the system of oppositions, and the whole of Part Two will read value off this coordinate vector.

Remark 3.9 (Why linearity matters mechanically)**.** *Saussure's point is that linearity is what makes the signifier* chainable*: elements succeed one another, and complex signs are built by concatenation. The algebra preserves exactly this: the GF(2) span builds complex nodes by*

linear combination of basis vectors, and materialization returns the chain. The one thing the lattice forgets—linear order*—is the one thing the door restores, which is why the sign relation needs both morphisms and not one [8]. Even the forgetting is not absolute: the windowed n-gram encoding used for ordered streams keeps* micro-context*—each n-gram is a small local order—and materialization can reassemble the original order from these micro-contexts by de Bruijn-style assembly, exactly as a de Bruijn graph reconstructs a sequence from its n-grams [7].*

4 Chapter II: Immutability and mutability of the sign

Chapter II of Part One states an apparent paradox: the sign is at once *immutable*—the community cannot change it—and *mutable*—time changes it continuously. Saussure gives four reasons for immutability, then derives mutability from the same principle of arbitrariness that seemed to guarantee immutability. The algebra resolves the paradox by separating two layers that the text keeps fused.

4.1 Immutability

The signifier "is imposed rather than freely chosen"; "the community itself cannot control so much as a single word; it is bound to the existing language" [2, Part I, ch. II, §2]. Saussure's four reasons are: (1) the arbitrariness of the sign, which gives no rational ground for preferring one signifier over another; (2) the multiplicity of signs necessary for any language; (3) the complexity of the system; (4) collective inertia toward innovation.

Reading 4.1 (Immutability). Immutability is the IICA contract, read at the level of the stored node. No operation of the system rewrites a node once written: ingestion is union, the commit store appends, and the lattice has no deletion primitive. The four Saussurean reasons map as follows.

Proposition 4.2 (Join monotonicity; no deletion)**.** *Every operation that writes the plane is a join. For all* $A, B \in \mathcal{B}$*,*

$$A \subseteq A \cup B,$$

so no legal write removes a bit once it is set. Individual and community alike can only accumulate *and* re-frame*; neither can rewrite a single stored sign. Collective inertia is a theorem of the algebra, not a sociological hope.*

Saussure's reason for immutability	HLLSet counterpart
(1) Arbitrariness gives no rational ground for preferring another signifier	Theorem 3.4: any re-coordination is isomorphic, so no node is a "better" name than another; there is nothing to optimize and therefore no individual reason to change
(2) Multiplicity of signs	$\|\mathcal{B}\| = 2^{32,768}$ nodes; the system is too large for any speaker to re-coordinate, and every re-coordination would have to be global to be consistent
(3) Complexity of the system	the frame: changing one coordinate (one basis pivot) re-values all nodes (generation increment), so the cost of change is systemic, not local
(4) Collective inertia	Proposition 4.2: joins only add; the stored plane is append-only; inertia is monotonicity

Table 1: The four Saussurean reasons for immutability, re-founded in the algebra.

4.2 Mutability

"Time changes all things; there is no reason why language should escape this universal law" [2, Part I, ch. II, §3]. The signifier and the signified shift their relationship: a language "is radically defenceless against the factors which constantly tend to shift the relationship" between the two faces, and this, Saussure adds, is a consequence of the very arbitrariness of the sign. Crucially, the shift is not made by speakers: "the language changes, but the speakers do not change it."

Reading 4.3 (Mutability). Mutability is the *generation mechanism* of the basis frames. The IICA bond (μ, ρ) is fixed, but the frame around every sign—the system of oppositions that gives the sign its value—is recomputed when the spill of new observations demands it. The value of each sign changes; the sign itself, as a stored node and as a content-addressable bond, does not.

Proposition 4.4 (Mutability as generation change)**.** *Let F be a basis frame with generation γ, and let a new observation Y arrive whose residual against F is non-empty,* $\mathrm{res}_F(Y) \neq \varnothing$ *(in particular, this holds when the spill $s = |Y \setminus \mathrm{cover}(F)|$ is positive). Recomputing the RREF basis produces a new frame F' with generation $\gamma + 1$. For an unchanged node $X \in \langle F \rangle$, the hard key generally changes, $c \mapsto c'$, although X itself is untouched (IICA) and its content is untouched (content addressability). The* value *of the sign has changed while the* sign *has not.*

4.3 The paradox resolved: two layers

Observation 4.5 (Immutability and mutability live on different layers). Saussure's text states both properties of the sign without separating the layers on which they operate. The algebra forces the separation, and the paradox dissolves:

1. **The contract layer** (IICA: the morphism μ, the stored node, the content-addressable bond) is immutable—by construction, by Proposition 4.2.
2. **The value layer** (the frame $F(t)$, the generation $\gamma(t)$, the hard and soft keys, the materialization door of each host) is mutable—by construction, by Proposition 4.4.

"The language changes although the speakers cannot change it" is then not a paradox but a type distinction: speakers can write only to the contract layer (by joining tokens), never to a node directly; the value layer changes globally whenever the frame pivots, and it is precisely this global, uncommanded change that is experienced as the drift of the language. The whole discussion is one instance of the fundamental equation of the state machine, $H(t) = (S(t), H(t-1), D, R, N)$ [7]: the commit stack has exactly these two layers—the emerging state and the committed base—and change is never overwriting; the new substitutes for the old by departing from it, while the old remains recorded.

Remark 4.6 (The defencelessness of the system)**.** *Saussure's "radically defenceless" now has a precise sense. The system has no guard that prevents the frame from pivoting, because the pivot is the normal response to spill—it is how the system learns. Defencelessness against diachronic drift is the price of openness to novelty; both are the same generation mechanism [7].*

5 Chapter III: Static and evolutionary linguistics

5.1 The two axes

Saussure distinguishes two axes of study. The *axis of simultaneities* (AB) concerns relations among coexisting terms, excluding time; the *axis of successions* (CD) concerns the same terms together with their changes, one at a time [2, Part I, ch. III, §1]. The distinction is "absolute and allows no compromise": synchronic linguistics deals with systems, diachronic linguistics with isolated events.

Reading 5.1 (The two axes). The axis of simultaneities is a *fiber* of the system at one time; the axis of successions is the *trajectory* across fibers.

Definition 5.2 (Synchronic fiber and diachronic trajectory)**.** *Fix a time t and a basis frame $F(t)$ of generation $\gamma(t)$. The* ***synchronic structure*** *at t is the fiber*

$$\mathsf{Syn}(t) \;=\; (\mathcal{B}, F(t), \gamma(t)),$$

the Boolean lattice together with its current frame: all nodes, their hard and soft keys, containment, complement, and the BSS readings among coexisting nodes. The ***diachronic structure*** *is the indexed family*

$$\mathsf{Dia} \;=\; \{\mathsf{Syn}(t) : t \in T\}$$

together with the recorded transitions—commits, departed/retained/new sets, generation increments—between consecutive fibers.

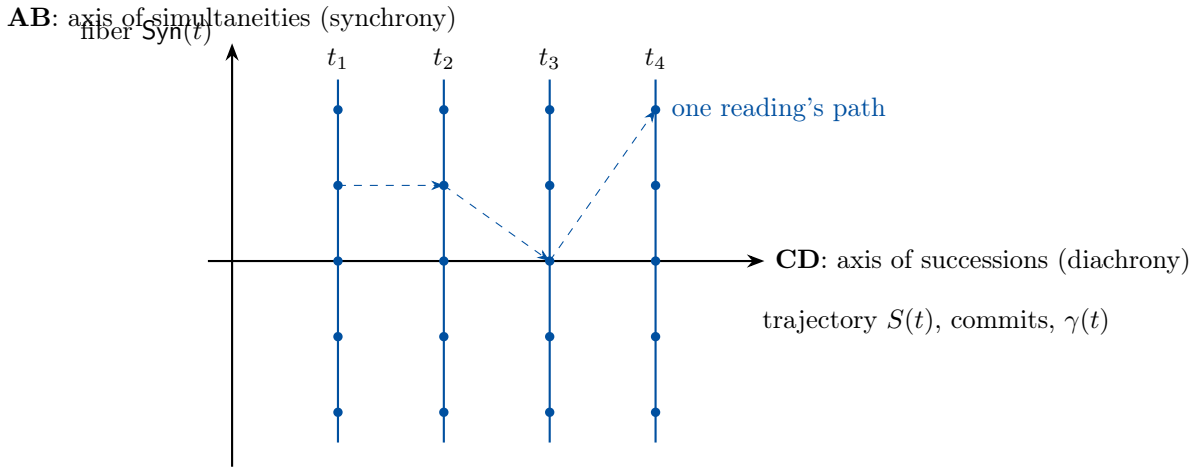


Figure 1: The two Saussurean axes as a fibration: each vertical fiber $\mathsf{Syn}(t)$ is a synchronic system (coexisting nodes, one frame, one generation); the dashed path is a diachronic reading across fibers.

5.2 The two kinds of law

Saussure's sharpest formulation in Chapter III is the asymmetry between the two kinds of law: a *synchronic* law is *general but not imperative*; a *diachronic* law is *imperative but not general* [2, Part I, ch. III, §3].

Theorem 5.3 (The two kinds of law, re-founded)**.** *(i)* Synchronic laws *are the identities of the Boolean lattice and ring in a fixed fiber $\mathsf{Syn}(t)$: for all nodes A, B, C of the fiber,*

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C), \qquad A \Delta A = \varnothing, \qquad \mathrm{cov}(X) \in [0,1],$$

together with the uniqueness of the hard key in the frame. These hold for every *node of the fiber—general—but they force no particular state to occur—not imperative. (ii)* Diachronic laws *are the recorded transitions: the commit $S(t) \to S(t{+}1)$ with its new and retained sets $N = S(t{+}1) \setminus S(t)$ and $R = S(t{+}1) \cap S(t)$, the pivot $F \to F'$, the generation increment $\gamma \to \gamma + 1$. Each is* imperative*—it happened, and the commit DAG records it—but* not general*—no identity forces it, and it holds of that transition alone.*

5.3 Panchronicity, and Saussure's warning

Saussure considers a third possibility, the *panchronic* viewpoint: general principles that hold independently of concrete facts. His warning is that panchronic facts, precisely because they are independent of concrete languages, never get to grips with the specific facts of any language [2, Part I, ch. III, §4].

Observation 5.4 (The panchronic layer is the ring axioms). The Boolean ring axioms are as follows: commutativity and associativity of both operations, distributivity of $\cap$ over Δ, idempotence

$$A \cap A = A,$$

and characteristic two

$$A \Delta A = \varnothing.$$

They hold in every fiber, in every generation, for every host language. They are the panchronic layer of the system. Saussure's warning translates exactly: these axioms are *true of everything* and therefore say nothing about any particular linguistic fact. The moment one asks a linguistic question—what is the value of this sign? how did this form change?—one leaves the panchronic layer and enters a fiber or a transition. The exegesis is faithful in this precise sense: it locates Saussure's panchronicity, not as an afterthought, but as the axioms of the algebra.

5.4 The chess analogy

Saussure's comparison: a game of chess is an artificial realization of what language presents in a natural form. A position is a synchronic state; a move is a diachronic event; and "the value of the pieces depends on their position on the chessboard, just as in the language each term has its value through its opposition to all the other terms" [2, Part I, ch. III, §5]. A move changes one piece, yet it can revolutionize the whole position; in language, likewise, a change is local but its consequences are systemic.

Reading 5.5 (The chess analogy). The board is the plane; a position is a synchronic fiber; a move is a commit.

Chess (Saussure)	Language (Saussure)	HLLSet Algebra
the board	the language as a system	the plane $\mathbb{B}$, fixed and shared
a position	a synchronic state	the fiber $\mathsf{Syn}(t) = (\mathcal{B}, F(t), \gamma(t))$
the value of a piece depends on its position	the value of a term depends on its oppositions	the hard key c and soft keys w_i relative to $F(t)$
a move changes one piece	a change is local	one commit; one token event joins one node
a move can revolutionize the whole position	a change has systemic consequences	a basis pivot increments γ and re-coordinates every cached reading [7]
the laws of chess outlive all games	panchronic principles	the Boolean ring axioms (Observation 5.4)

Table 2: The chess analogy, completed by the algebra.

Remark 5.6 (Why the analogy is exact, not illustrative)**.** *In chess the value of a piece is not written on the piece; it is computed from the position. In HLLSet Algebra the value of a node is not written on the node; it is computed from the frame. The algebra is the first setting in which Saussure's analogy stops being an analogy and becomes the definition of a datatype: value is a function of (node, frame), and nothing else.*

6 The dictionary of Part One

Table 3 collects the complete dictionary of Part One. Each row is either a theorem, a proposition, or a definition of this paper; nothing in the table is left as a suggestive parallel.

Saussure (Part One)	In the *Course*	In HLLSet Algebra
linguistic sign	two-sided psychological entity	(μ, ρ) with faces (node, value): Definition 2.2, Reading 3.1
signifier (sound pattern)	the acoustic image	the node $A \subseteq \mathbb{B}$: a bit-set on the one-dimensional line
signified (concept)	the concept	the differential value: hard key c, complement $\overline{A}$, materialized context
bond between the faces	the arbitrary link	the morphism pair (μ, ρ); round-trip guarded by coverage
Principle I: arbitrariness	unmotivated, yet conventional	Theorem 3.4 (coordinate freedom); Corollary 3.5 (fixed by IICA)
Principle II: linearity	the signifier is a line in time	Proposition 3.8: medium line + GF(2)-linearity
immutability of the sign	the community cannot change it	Proposition 4.2 (join monotonicity; no deletion); Table 1
mutability of the sign	time shifts the sign	Proposition 4.4 (generation change); Observation 4.5 (two layers)
axis of simultaneities	AB: coexisting terms	the fiber $\mathsf{Syn}(t)$, Definition 5.2
axis of successions	CD: terms and their changes	the trajectory Dia and the commit DAG, Definition 5.2
synchronic law	general but not imperative	lattice/ring identities in a fiber, Theorem 5.3(i)
diachronic law	imperative but not general	recorded transitions, Theorem 5.3(ii)
panchronic viewpoint	general, independent of facts	the Boolean ring axioms, Observation 5.4
chess analogy	value depends on position	value as a function of (node, frame), Table 2

Table 3: The complete dictionary of Part One of the *Course* into HLLSet Algebra.

7 What the reversal changes

The five EWM papers read Saussure into the algebra; this paper reads the algebra into Saussure. The reversal is not a symmetry of exposition. It changes five things.

1. **From analogy to construction.** In the EWM direction, Saussurean terms were lenses: they helped *see* the algebra as linguistics. In the reverse direction, the terms are *requirements*: the algebra must satisfy them or fail. Part One satisfies them: every principle above is a theorem or an invariant.
2. **Ambiguity becomes a type error.** Philosophical commentary can live with the ambiguity between the two layers of immutability/mutability. A datatype cannot. The algebra forces Observation 4.5, and the forced distinction is arguably what Saussure's text was reaching for.
3. **The text becomes testable.** Each Reading can be checked by running the reference implementation: ingest a corpus, pivot a frame, and verify that stored nodes are untouched while hard keys change. The exegesis is falsifiable in the laboratory, not only in the seminar.
4. **The direction of verification reverses.** Previously Saussure validated the EWM programme; the risk was that the *Course* was being used decoratively. Now the EWM programme validates Saussure; the risk is the opposite one, that the algebra is being used to settle questions the text deliberately left open. We therefore record, at each step, whether the algebra *satisfies* a principle or *disambiguates* it.
5. **A residual is a finding, not a failure.** Where the algebra has no counterpart for a Saussurean claim, that is itself a result: it locates the claim outside the computable core. Part One leaves, so far, one such residual: the psychological character of the two faces, which we replaced by structural two-sidedness without pretending the replacement is lossless.

8 Conclusion and outlook to Part II

Part One of the *Course in General Linguistics* has been re-founded, in the reverse direction, as a sequence of definitions, theorems, and invariants of HLLSet Algebra. The sign is (μ, ρ); arbitrariness is coordinate freedom; linearity is the double line of bit positions and GF(2)-span; immutability is join monotonicity; mutability is generation change; synchrony is a fiber, diachrony a trajectory, panchrony the ring axioms; and the chess analogy is the definition of value as a function of (node, frame). Nothing in the reading is a metaphor, and everything is checkable against the running system.

Part II will take up where Part One deliberately stops: the differential theory of value. Saussure's "in the language itself there are only differences without positive terms" will be read as the statement that the value of a node is its hard key in the current frame, that the frame itself is a system of distinctive oppositions, and that the two axes of Part One—simultaneities and successions—become, in Part Two, the paradigmatic and syntagmatic relations of the lattice. The reversal of direction will then be complete in both directions: the *Course* interpreted by the algebra, and the algebra interpreted by the *Course*, as two readings of one construction.

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